SPARK Academy Multi-Project Hackathon
Code Hack & Team Exercise Tasks

Instructions

Use the **Code Hack Sessions** to implement, adapt, and run your selected project model using the assigned computing environment, such as Compute Canada, Kaggle, Colab, or a local/server environment approved by your site.

Each team should work toward producing a reproducible baseline result for its assigned project. Depending on the project type, this may involve classification, segmentation, object detection, image reconstruction, uncertainty estimation, explainability, or referral-triage modeling.

Use the **Team Exercise Sessions** to prepare your capstone presentation, document implementation progress, troubleshoot code, refine experiments, or continue model training and evaluation.

Capstone Project Presentation

Each team will deliver a **5-minute presentation** summarizing the team's model implementation, performance, challenges, and lessons learned.

For hybrid and virtual sites, Code Hack and Team Exercise Sessions will take place through the program platform in **Google Meet**. Teams should use their assigned virtual meeting room or site-specific communication channel.

Daily Activities:

<table>
<tr><th>Day</th><th>Code Hack Task and Team Exercise</th></tr>
<tr><td>Monday</td><td>
Project Planning, Dataset Review, and Model Selection

Review the assigned project brief, dataset description, clinical motivation, and project objectives. Review the recommended method papers and GitHub repositories relevant to your project type. These may include models for classification, segmentation, object detection, image reconstruction, uncertainty estimation, explainability, or generative enhancement.
Choose an initial baseline model or workflow appropriate for your project. Examples include CNNs, U-Net, nnU-Net, UNETR, SwinUNETR, YOLO/Faster R-CNN, Vision Transformers, reconstruction networks, uncertainty-aware models, or saliency/explainability pipelines.
Review available starter code in Kaggle, Colab, GitHub, or the assigned compute environment.

Deliverable: 1-page report

Submit a summary outlining:
Project title and clinical/research motivation
Dataset and label structure
Chosen model or baseline workflow
Model architecture or pipeline summary, preferably in a table
Novelty or rationale for the selected method
Expected training parameters and optimization approach
Anticipated implementation challenges

Submit the mid-program evaluation, if requested by the organizing team.
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<td>Tuesday</td>
<td>Data Exploration and Preprocessing

Perform exploratory data analysis. Understand the dataset format, label structure, class balance, image dimensions, metadata, and available annotations.
Depending on the project, inspect formats such as DICOM, NIfTI, JPG/PNG, microscopy images, ultrasound images, mammograms, chest X-rays, k-space files, masks, bounding boxes, BI-RADS descriptors, or clinical labels.
Preprocess the data for model training. This may include resizing, normalization, cropping, lung/breast/cervix/tumor region preparation, mask conversion, bounding-box parsing, train-validation-test splitting, augmentation, or k-space preprocessing.
Transfer starter code from notebook format to reproducible Python scripts where needed. Begin running the baseline model in the assigned compute environment.

Exercise: 1-page report
Outline:

Dataset structure and preprocessing steps completed
Data quality issues identified
Class imbalance or annotation issues
Any challenges faced in preparing or running the model

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<td>Wednesday</td>
<td>Model Implementation and Initial Training

Train the chosen baseline model on the assigned project dataset.

Ensure that the model objective matches the project endpoint. For example:

★ Classification projects should define class-wise metrics and clinically important errors.
★ Segmentation projects should evaluate mask quality and boundary reliability.
★ Detection projects should evaluate localization, false positives, and false negatives.
★ Reconstruction projects should evaluate image fidelity and artifact behavior.
★ Triage projects should evaluate referral groups, uncertainty, and safety thresholds.
★ Explainability projects should evaluate whether attention is anatomically plausible.

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Track training logs, validation curves, errors, and hardware/runtime issues.
Exercise: 1-page report
Submit:

Initial or preliminary training results
Current performance metrics
Training challenges or code issues
Any early signs of model failure, overfitting, poor generalization, or unsafe predictions

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<tr><td>Thursday</td><td>
Model Validation, Evaluation, and Failure Analysis

Continue training and validate the model on the assigned validation data.
Evaluate model performance using metrics appropriate to the project. Examples include accuracy, sensitivity, specificity, F1-score, AUROC, AUPRC, Dice score, Hausdorff distance, mAP, IoU, SSIM, PSNR, NRMSE, calibration error, uncertainty measures, false-negative rate, or subgroup performance.
Compare your results with the baseline paper, published result, reference implementation, or expected performance where available.

Perform project-specific failure analysis. Examples include:

★ Missed TB cases
★ Dense-breast mammography failures
★ Type 3 cervix misclassification
★ Malaria WBC false positives
★ Ultrasound boundary errors
★ MRI missing-modality failure
★ Low-field MRI reconstruction artifacts
★ Non-anatomical saliency or shortcut learning

Exercise: 1-page report
Submit:

Validation results
Key model deficiencies
Error or failure-mode analysis
Strategies to improve the model
Whether the model output appears clinically or scientifically trustworthy

Each team member should complete the mid-training survey
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<td>Friday</td>
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Final Evaluation, Optimization, and Capstone Preparation

Finalize remaining model evaluation, plots, tables, confusion matrices, segmentation overlays, detection examples, reconstruction comparisons, uncertainty outputs, or saliency maps.
Prepare a concise interpretation of what worked, what failed, and what would be improved with more time.
Emphasize safe and honest reporting. Teams should not overclaim clinical diagnosis, lesion localization, or deployment readiness unless the dataset and labels support those claims.

Exercise: Prepare the 5-minute Capstone Presentation
Capstone Presentation Requirements:

★ 5 minutes maximum
★ No more than 8 slides

Present a summary of:

Project title and motivation
Dataset used and label/annotation structure
Why the model or method was chosen
Model architecture or workflow
Training and preprocessing strategy
Performance results and whether expected performance was replicated
Observed or suspected model deficiencies
Challenges faced during implementation
Safety, uncertainty, explainability, or deployment considerations
Proposed next-step model or project extension

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Recommended Capstone Slide Structure

Slide 1: Project title, team members, disease/domain, and clinical motivation

Slide 2: Dataset description, labels, annotations, and task framing

Slide 3: Model architecture or computational workflow

Slide 4: Preprocessing and training strategy

Slide 5: Main results and performance metrics

Slide 6: Failure analysis, uncertainty, subgroup analysis, or explainability findings

Slide 7: Challenges faced and lessons learned

Slide 8: Proposed next step, improved model, or future challenge submission

General Team Expectations

Each team should finish the week with:

- **Project summary:** Clear research question, dataset, and endpoint
- **Working code:** Reproducible baseline implementation or adapted starter code
- **Data pipeline:** Documented preprocessing and train/validation split
- **Model results:** Initial performance metrics and visual outputs where relevant
- **Failure analysis:** Description of model limitations and unsafe failure modes
- **Capstone slides:** 5-minute presentation with no more than 8 slides
- **Next-step plan:** Proposed model improvement, extension, or future submission pathway